Deep Learning for AI

# Image Classification (Landscapes)

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# Introduction

- **Objective:** Perform an image classification task using deep learning models. Model is trained to identify different categories of objects in images.
- **Libraries:** “torch”, “torchvision”, “matplotlib”, “tqdm”
- **Dataset:** Intel Image Classification dataset
    - 6 classes (buildings,  forest, glacier, mountain, sea, street)
    - Images resized to 224x224
    - **Training set:** for model training,  **Validation set:** for performance evaluation

- **Random samples from Training data**
    - 25 random images.

- **Model:** ResNet-50

  ResNet-50 became a standard in image classification tasks and it was selected for this project as well because it is a deep CNN model that allows the model train deeper layers effectively thanks to the it's residual connections. This solves the general vanishing gradients issue, improves model accuracy.

- **Activation function:** ReLU
- **Optimizer:** Adam optimizer
- **Loss function:** CrossEntropy Loss function

- **Automatic Differentiation**

  During training the model, backpropagation is used to calculate gradients by autograd in PyTorch.

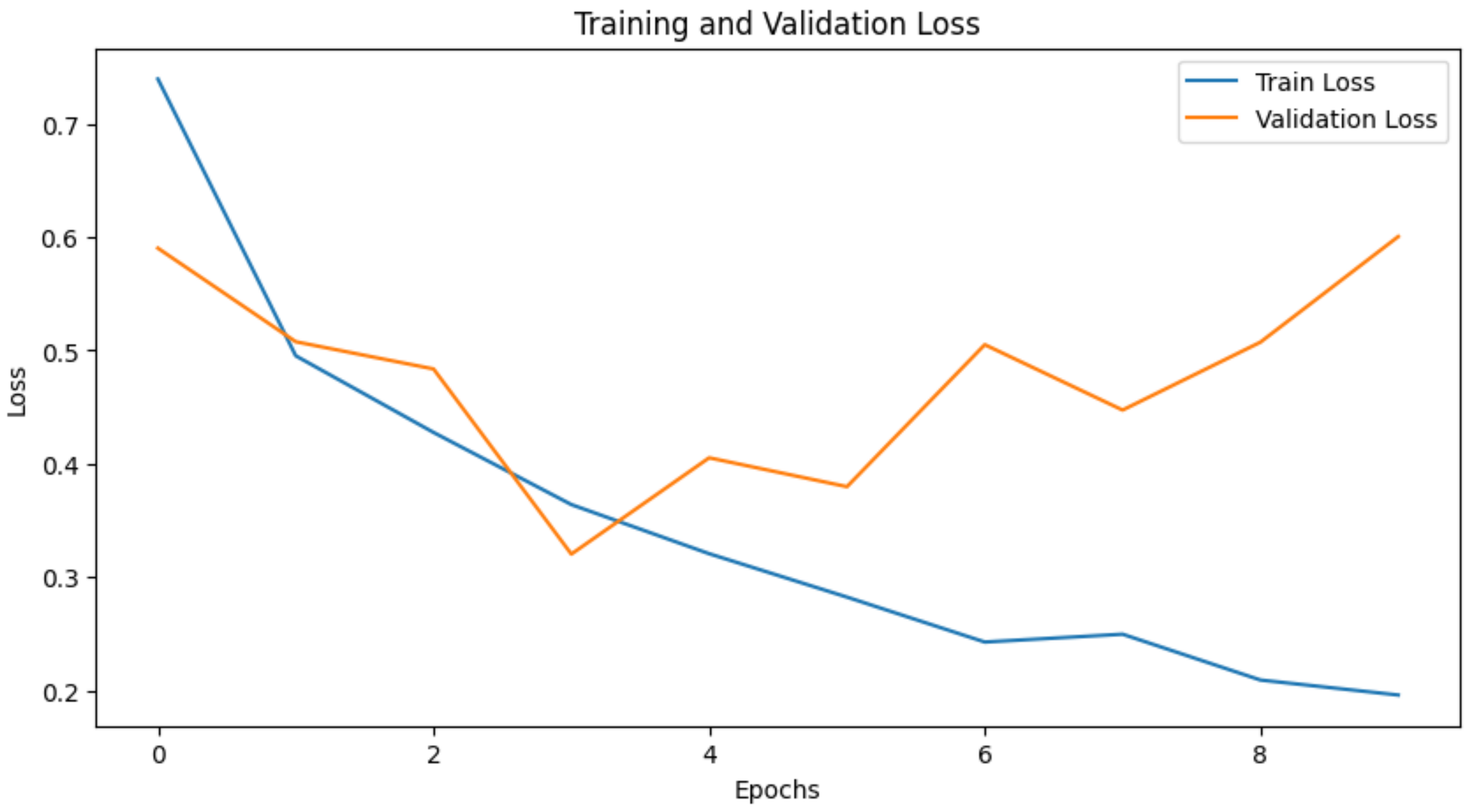
- **Model Training**

  10 epochs: training data pas passed through the model 10 times

  Batch size = 32: 32 images were processed in each training step.

  Learning rate is 0.001

```
Epoch 1/10: 100% 188/188 [01:03<00:00, 3.77it/s]
Epoch 1: Train Loss = 0.1270, Val Loss = 0.3720, Val Accuracy = 0.8788
Epoch 2/10: 100% 188/188 [01:02<00:00, 3.62it/s]
Epoch 2: Train Loss = 0.1342, Val Loss = 0.4020, Val Accuracy = 0.8679
Epoch 3/10: 100% 188/188 [01:02<00:00, 3.60it/s]
Epoch 3: Train Loss = 0.1431, Val Loss = 0.3765, Val Accuracy = 0.8762
Epoch 4/10: 100% 188/188 [01:02<00:00, 3.71it/s]
Epoch 4: Train Loss = 0.1305, Val Loss = 0.3756, Val Accuracy = 0.8779
Epoch 5/10: 100% 188/188 [01:02<00:00, 3.57it/s]
Epoch 5: Train Loss = 0.1380, Val Loss = 0.3970, Val Accuracy = 0.8742
Epoch 6/10: 100% 188/188 [01:02<00:00, 3.70it/s]
Epoch 6: Train Loss = 0.1291, Val Loss = 0.3898, Val Accuracy = 0.8721
Epoch 7/10: 100% 188/188 [01:02<00:00, 3.66it/s]
Epoch 7: Train Loss = 0.1422, Val Loss = 0.3721, Val Accuracy = 0.8758
Epoch 8/10: 100% 188/188 [01:01<00:00, 3.65it/s]
Epoch 8: Train Loss = 0.1383, Val Loss = 0.3819, Val Accuracy = 0.8729
Epoch 9/10: 100% 188/188 [01:03<00:00, 3.68it/s]
Epoch 9: Train Loss = 0.1291, Val Loss = 0.3856, Val Accuracy = 0.8729
Epoch 10/10: 100% 188/188 [01:03<00:00, 3.60it/s]
Epoch 10: Train Loss = 0.1317, Val Loss = 0.3922, Val Accuracy = 0.8717
```


Training and Validation Loss
Train Loss
Validation Loss
Loss
0.7
0.6
0.5
0.4
0.3
0.2
0
2
4
6
8
Epochs

- **Results and Performance Evaluation**
- During the training, accuracy and loss metrics are monitored over 10 epochs.
- Model’s performance was visualised with confusion matrix.

True label
Predicted label
buildings
forest
glacier
mountain
sea
street
272 0 0 1 6 121
0 389 0 2 0 9
1 3 221 84 60 31
3 10 13 336 21 17
2 5 1 13 364 15
8 2 0 2 5 383
350
300
250
200
150
100
50
0

- Misclassification Examples

Prediction: sea, True: buildings

Prediction: street, True: buildings

Prediction: street, True: buildings

Prediction: street, True: buildings

Prediction: street, True: buildings